

Cloud Operations Runbook

Scenario: High CPU and Memory Alarms on Virtual Machines

1. Purpose

This runbook provides a simple, standardized procedure to respond to high CPU and memory utilization alerts on virtual machines (VMs). The current remediation strategy is to restart the affected VM.

2. Scope

- Applies to all VMs monitored in the cloud environment
 - Triggered by monitoring alerts for:
 - CPU utilization above defined threshold
 - Memory utilization above defined threshold
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3. Prerequisites

- Access to cloud console (OCI / relevant cloud provider)
 - Permissions to view metrics and restart instances
 - Monitoring/alerting system access (OCI Monitoring, PagerDuty, etc.)
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4. Alert Trigger Conditions

Metric	Threshold	Duration
CPU Utilization	> 85%	5 minutes
Memory Utilization	> 90%	5 minutes

5. Procedure

Step 1: Restart VM (Primary Remediation)

- Notify stakeholders (if required)
- Perform VM restart via cloud console or CLI

OCI Example:

```
oci compute instance action --instance-id <instance_id> --action SOFTRESET
```

Step 5: Post-Restart Validation

- Confirm VM is back online
- Re-check CPU and memory metrics
- Validate application availability

Step 6: Close Alert

- Resolve/close alert in monitoring system
 - Document actions taken
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6. Escalation Criteria

Escalate to engineering team if:

- Issue persists after restart
 - VM becomes unresponsive post-restart
 - Alerts recur more than 3 times in 1 hour
 - Critical application impact is observed
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7. Risks and Considerations

- Restart may cause application downtime

- Possible data loss for in-memory workloads
 - Does not address root cause
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8. Future Improvements (Recommended)

- Implement auto-scaling instead of restart
 - Add process-level monitoring
 - Introduce automated remediation (Functions / Events)
 - Perform root cause analysis for recurring issues
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9. Runbook Owner

- Cloud Operations Team
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